Student #1: 15359 (X1=1; X2=5; X3=3; X4=5; X5=9)
Student #2: 12657 (X10=1; X9=2; X8=6; X7=5; X6=7)

	Calendar Inc	Horoscope Inc
Public IPv4 Network	203.153.151.128/25	203.012.065.0/25
Private IPv4 Network	172.29.012.0/23	172.22.072.0/23
Global IPv6 Network	2002:A153:BC59::/48	2002:A126:BC57::/48

Public IPv4 - Calender Inc

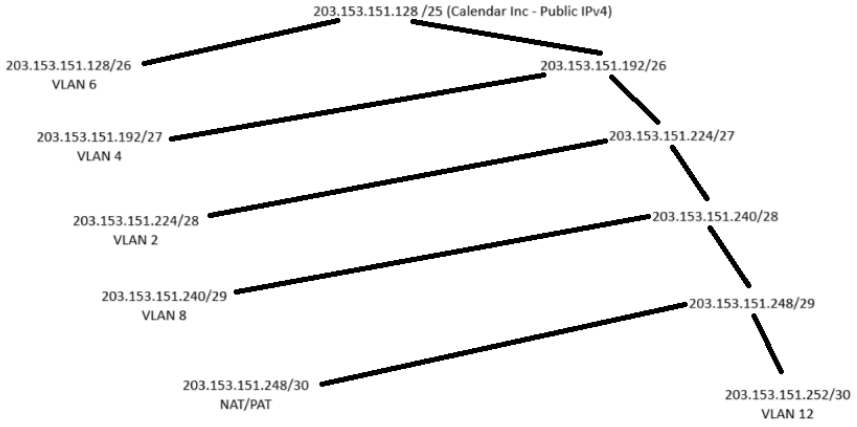
	Network Address	Mask	Broadcast Address	Range of Address	Gateway #1	Gateway #2
VLAN 2	203.153.151.224	28	203.153.151.239	203.153.151.226 to 203.153.151.238	203.153.151.225	N/A
VLAN 4	203.153.151.192	27	203.153.151.223	203.153.151.194 to 203.153.151.222	203.153.151.193	N/A
VLAN 6	203.153.151.128	26	203.153.151.191	203.153.151.130 to 203.153.151.190	N/A	203.153.151.129
VLAN 8	203.153.151.240	29	203.153.151.247	203.153.151.242 to 203.153.151.246	N/A	203.153.151.241
VLAN 12	203.153.151.252	30	203.153.151.255	203.153.151.254	203.153.151.253	N/A
NAT/PAT	203.153.151.248	30	203.153.151.251	203.153.151.249 to 203.153.151.250	N/A	N/A

The base public address for this assignment is 203.1X4X3.1X2X1.128/25, or 203.153.151.128/25 when replacing with the NºMec of Student #1. This means we will have a total of 128 addresses to use.

We will have to divide the network for the following requirements of each VLAN's:

- VLAN 2: 12 addresses;
- VLAN 4: 20 addresses;
- VLAN 6: 50 addresses;
- VLAN 8: 5 addresses;
- VLAN 12: 1 address;
- NAT/PAT: 2 addresses;

Broadcast: will always be the last address of a sub-network
Gateway: will be the first address available in the sub-network



Public IPv4 - Horoscope Inc

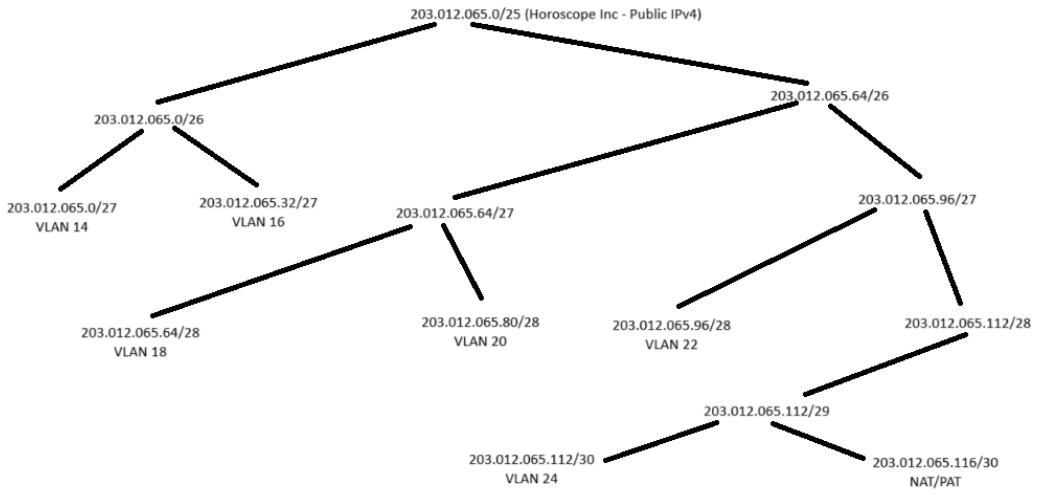
	Network Address	Mask	Broadcast Address	Range of Address	Gateway #1	Gateway #2
VLAN 14	203.012.065.0	27	203.012.065.31	203.012.065.2 to 203.012.065.30	N/A	203.012.065.1
VLAN 16	203.012.065.32	27	203.012.065.63	203.012.065.34 to 203.012.065.62	N/A	203.012.065.33
VLAN 18	203.012.065.64	28	203.012.065.79	203.012.065.66 to 203.012.065.78	203.012.065.65	N/A
VLAN 20	203.012.065.80	28	203.012.065.95	203.012.065.82 to 203.012.065.94	203.012.065.81	N/A
VLAN 22	203.012.065.96	28	203.012.065.111	203.012.065.98 to 203.012.065.110	203.012.065.97	N/A
VLAN 24	203.012.065.112	30	203.012.065.115	203.012.065.114	N/A	203.012.065.113
NAT/PAT	203.012.065.116	30	203.012.065.119	203.012.065.117 to 203.012.065.118	N/A	N/A

The base public address for this assignment is 203.0X10X9.0X8X7.0/25, or 203.012.065.0/25 when replacing with the NºMec of Student #2. This means we will have a total of 128 addresses to use.

We will have to divide the network for the following requirements of each VLAN's:

- VLAN 14: 28 addresses;
- VLAN 16: 27 addresses;
- VLAN 18: 13 addresses;
- VLAN 20: 10 addresses;
- VLAN 22: 7 addresses;
- VLAN 24: 1 address:
- NAT/PAT: 3 addresses;

Broadcast: will always be the last address of a sub-network
Gateway: will be the first address available in the sub-network



Private IPv4 - Calender Inc

	Network Address	Mask	Broadcast Address	Range of Address	Gateway #1	Gateway #2
VLAN 2	172.29.012.0	24	172.29.012.255	172.29.012.2 to 172.29.012.254	172.29.012.1	N/A
VLAN 4	172.29.013.0	25	172.29.013.127	172.29.013.2 to 172.29.013.126	172.29.013.1	N/A
VLAN 6	172.29.013.128	26	172.29.013.191	172.29.013.130 to 172.29.013.190	N/A	172.29.013.129
VLAN 8	172.29.013.192	27	172.29.013.223	172.29.013.194 to 172.29.013.222	N/A	172.29.013.193
VLAN 12	172.29.013.224	30	172.29.013.227	172.29.013.226 to 172.29.013.226	172.29.013.225	N/A
ESW2	172.29.013.228	30	172.29.013.231	172.29.013.229 to 172.29.013.230	N/A	N/A

The base public address for this assignment is 172.2X5.0X12.0/23, or 172.29.012.0/23 when replacing with the NºMec of Student #1. This means we will have a total of 512 addresses to use.

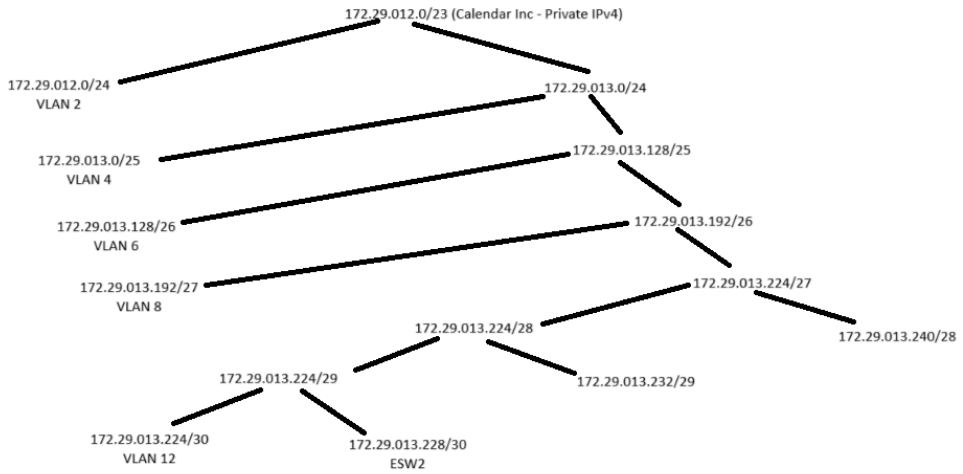
We will have to divide the network for the following requirements of each VLAN's:

- VLAN 2: 200 addresses;
- VLAN 4: 110 addresses;
- VLAN 6: 50 addresses;
- VLAN 8: 25 addresses;
- VLAN 12: 0 addresses;

And we will be using 172.29.013.228 for ESW2.

Broadcast: will always be the last address of a sub-network.

We will have unused addresses in 172.29.013.232/29 and 172.29.013.240/28.



Private IPv4 - Horoscope Inc

	Network Address	Mask	Broadcast Address	Range of Address	Gateway #1	Gateway #2
VLAN 14	172.22.073.128	27	172.22.073.159	172.22.073.130 to 172.22.073.158	N/A	172.22.073.129
VLAN 16	172.22.073.0	26	172.22.073.63	172.22.073.2 to 172.22.073.62	N/A	172.22.073.1
VLAN 18	172.22.073.64	26	172.22.073.127	172.22.073.66 to 172.22.073.126	172.22.073.65	N/A
VLAN 20	172.22.073.160	28	172.22.073.175	172.22.073.162 to 172.22.073.174	172.22.073.161	N/A
VLAN 22	172.22.072.0	24	172.22.072.255	172.22.072.2 to 172.22.072.254	172.22.072.1	N/A
VLAN 24	172.22.073.176	30	172.22.073.179	172.22.073.178	N/A	172.22.073.177
ESW1	172.22.073.180	30	172.22.073.183	172.22.073.181 to 172.22.073.182	N/A	N/A

The base public address for this assignment is 172.2X9.0X62.0/23, or 172.22.072.0/23 when replacing with the NºMec of Student #2. This means we will have a total of 512 addresses to use.

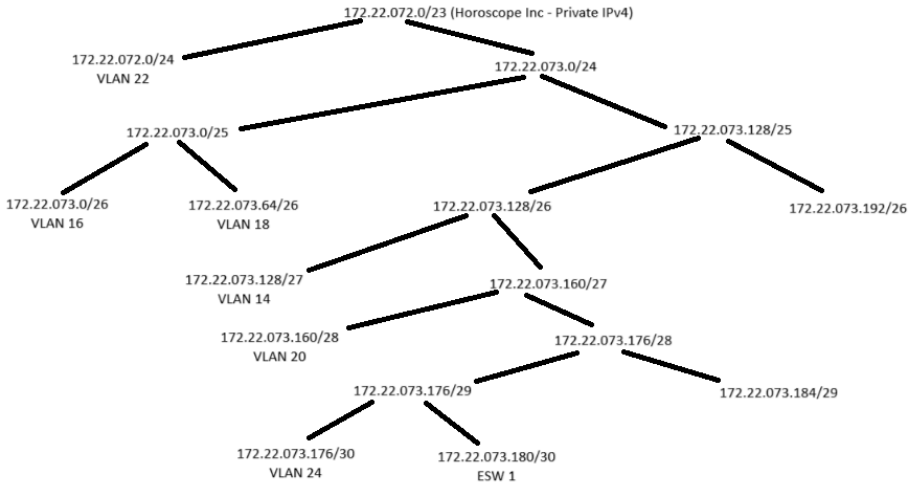
We will have to divide the network for the following requirements of each VLAN's:

- VLAN 14: 25 addresses;
- VLAN 16: 55 addresses;
- VLAN 18: 57 addresses;
- VLAN 20: 10 addresses;
- VLAN 22: 155 addresses;
- VLAN 24: 0 addresses;

And we will be using 172.22.073.180 for ESW1.

Broadcast: will always be the last address of a sub-network.

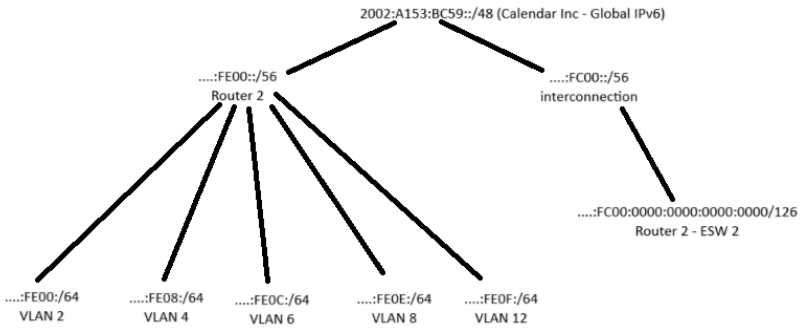
We will have unused addresses in 172.22.073.184/29 and 172.22.073.192/26.



Global IPv6 - Calendar Inc

	Network Address	Mask	Range of Address	Gateway #1	Gateway #2
VLAN 2	GP:FE00::	64	GP:FE00:0002 to GP:FE00:FFFF:FFFF	GP:FE00:0001	N/A
VLAN 4	GP:FE08::	64	GP:FE08:0002 to GP:FE08:FFFF:FFFF	GP:FE08:0001	N/A
VLAN 6	GP:FE0C::	64	GP:FE0C:0002 to GP:FE0C:FFFF:FFFF	N/A	GP:FE0C:0001
VLAN 8	GP:FE0E::	64	GP:FE0E:0002 to GP:FE0E:FFFF:FFFF	N/A	GP:FE0E:0001
VLAN 12	GP:FE0F::	64	GP:FE0F:0002 to GP:FE0F:FFFF:FFFF	GP:FE0F:0001	N/A
R2 to ESW2	GP:FC00:0000:0000:0000:0000	126	GP:FC00::0001 to GP:FC00::0002	N/A	N/A

Global Prefix: GP = 2002:A153:BC59



Global IPv6 - Horoscope Inc

	Network Address	Mask	Range of Address	Gateway #1	Gateway #2
VLAN 14	GP:FF00::	64	GP:FF00:0002 to GP:FF00:FFFF:FFFF	N/A	GP:FF00:0001
VLAN 16	GP:FF08::	64	GP:FF08:0002 to GP:FF08:FFFF:FFFF	N/A	GP:FF08:0001
VLAN 18	GP:FF0C::	64	GP:FF0C:0002 to GP:FF0C:FFFF:FFFF	GP:FF0C:0001	N/A
VLAN 20	GP:FF0E	64	GP:FF0E:0002 to GP:FF0E:FFFF:FFFF	GP:FF0E:0001	N/A
VLAN 22	GP:FF0F	64	GP:FF0F:0002 to GP:FF0F:FFFF:FFFF	GP:FF0F:0001	N/A
VLAN 24	GP:FF10	64	GP:FF10:0002 to GP:FF10:FFFF:FFFF	N/A	GP:FF10:0001
R1 to ESW1	GP:FC00:0000:0000:0000:0000	126	GP:FC00::0001 to GP:FC00::0002	N/A	N/A

Global Prefix: GP = 2002:A126:BC57

